SQL ASSIGNMENT 1: DATABASE SETUP & FUNDAMENTALS

Subject: Database Management Systems (SQL) **Topic:** DDL, DML & Database Concepts **Status:** Completed

TASK 1: DATABASE CREATION

Install MySQL/PostgreSQL and create a new database named `music_streaming_app` using command line or GUI tool.

```
-- Command Line Syntax to create and connect to database CREATE DATABASE music_streaming_app; -- Switch to the newly created database USE music_streaming_app; -- MySQL syntax -- \c music_streaming_app -- PostgreSQL CLI syntax
```

TASK 2: TABLE SCHEMA DEFINITION

Inside `music_streaming_app`, create a table called `playlists` with columns: `playlist_id` (integer, primary key), `name` (varchar), and `created_by` (varchar).

```
CREATE TABLE playlists ( playlist_id INT PRIMARY KEY, name VARCHAR(100) NOT NULL, created_by VARCHAR(50) NOT NULL );
```

TASK 3: SAMPLE DATA INSERTION

Insert three sample rows representing playlists ('Bollywood Hits', 'Chill Vibes', 'Workout Mix'), each created by a different user.

```
INSERT INTO playlists (playlist_id, name, created_by) VALUES (1, 'Bollywood Hits', 'Amit'), (2, 'Chill Vibes', 'Sneha'), (3, 'Workout Mix', 'Rahul');
```

TASK 4: DATA FILTERING WITH WHERE

Write a SQL SELECT query to display all playlists created by the user 'Amit' from the `playlists` table.

```
SELECT * FROM playlists WHERE created_by = 'Amit';
```

TASK 5: SQL CONCEPTS EXPLANATION (ZOMATO ANALOGY)

Explanation of the difference between a Table, Row, and Column using a Zomato food delivery app example:

Imagine the database of a food delivery app like **Zomato**:

- **Table (The Entire Spreadsheet/Container):** A Table is a structured collection of related data stored in rows and columns. In Zomato, an `Orders` table holds all information about every food order ever placed.
- **Column (The Attribute/Field):** A Column represents a specific category or type of information stored vertically across all records. In Zomato's `Orders` table, individual columns would be `order_id`, `restaurant_name`, `food_item`, and `total_price`.
- **Row (The Single Record/Tuple):** A Row represents a single, complete entry stored horizontally containing data for all columns. For instance, one row in Zomato would store: `[Order #5092 | "Truffles" | "Chicken Burger" | ₹350]`.